

Finding the Interquartile Range (IQR) of Data

6.SP.5c

Thank you for your purchase!

This product contains the following:

1. IQR Cloze notes and Guided Practice
2. 20 Task Cards
3. Answer Sheet
4. Answer Key

This product is designed to help students master the following:

1. Find the IQR of an even or odd set of data
2. Determine the IQR of data in line plots, bar graphs, and box and whisker plots

These cards are great for centers and/or partner activities. Print and laminate for long-term enjoyment!

Font: Vanessa Bays @ <http://bythebutterfly.com>

Background: <http://www.etsy.com/shop/GreatGraphics>

<http://www.weatherbase.com/weather/state.php3?c=US&s=&countryname=United-States> (precipitation data)

<http://www.weatherbase.com/weather/city.php3?c=US&s=NC&statename=North-Carolina-United-States-of-America> (NC temperature data)

When you find the Interquartile Range (IQR) of a set of data, you need _____ data points.

1. The _____ is the middle of the data. Make sure you put your data in _____ from _____ to greatest first! The median helps you cut your data into the lower and upper quartiles
2. The _____ quartile (or quartile 1) is found by finding the _____ of the lower half of the data.
3. The _____ quartile (or quartile 3) is found by finding the _____ of the upper half of the data.

Finding the IQR with an Odd # of Data Points

1. Put the numbers in _____ from least to greatest and find the _____ of the data _____ it.
2. Find the _____ of the lower quartile (the numbers in front of your median)
3. Find the _____ of the upper quartile (the numbers behind your median)

(Hint: If a quartile contains an even # of data points, _____ the two middle numbers and _____ by 2 to get your median)

4. _____ the lower quartile from the upper quartile ($Q3 - Q1$).

Finding the IQR with an Even # of Data Points

1. Put the numbers in _____ from least to greatest.
2. Break your data into _____ equal groups. Draw a line between the two groups.
3. Find the _____ of the front group. (lower quartile)
4. Find the _____ of the back group. (upper quartile)
5. _____ the lower quartile from the upper quartile ($Q3 - Q1$).

Let's Try it! – Remember: Median, Median, Median, Subtract!

Finding the IQR with an Odd # of Data Points

A rescue agency recorded the number of pets adopted each month:

12, 18, 22, 15, 29, 12, 8, 17, 16

What is the IQR of the # of pets adopted at the rescue agency?

1. Put the numbers in order.
2. What is your median? Circle it.
3. Find the median of the numbers in front of your median (Q1). There are 4 numbers, but the two in the middle are the same, so that is your median, or Q1. $Q1 = \underline{\hspace{2cm}}$
4. Find the median of the numbers behind your median (Q3). There are four numbers. The two in the middle are not the same, so add them up and divide by 2. $Q3 = \underline{\hspace{2cm}}$
5. Now $Q3 - Q1 =$

Let's Try Another!

Find the IQR of the data below:

23, 18, 25, 21, 21, 17, 12

Lower Quartile:

Upper Quartile:

Interquartile Range:

Finding the IQR with an Even # of Data Points

A car rental agency rented a certain number of cars each day::

25, 12, 15, 11, 18, 20

What is the IQR of the # of cars rented at the agency?

1. Put the numbers in order.
2. Cut the #s into two equal groups. Draw a line to separate them.
3. Find the median of the numbers in the front half – this is Q1. $Q1 = \underline{\hspace{2cm}}$
4. Find the median of the numbers in the back half – this is Q3. $Q3 = \underline{\hspace{2cm}}$
5. Now $Q3 - Q1 =$

Let's Try Another!

Find the IQR of the data below:

21, 17, 15, 23, 21, 17, 12, 9

Lower Quartile:

Upper Quartile:

Interquartile Range:

Use the data set below to identify each of the following:

23, 18, 25, 21, 21

Lower Quartile:

Upper Quartile:

Interquartile Range:

1

Use the data set below to identify each of the following:

15, 24, 17, 17, 21, 25, 15

Lower Quartile:

Upper Quartile:

Interquartile Range:

2

Use the data set below to identify each of the following:

23, 17, 22, 23, 21, 18

Lower Quartile:

Upper Quartile:

Interquartile Range:

3

Use the data set below to identify each of the following:

4.5, 3.9, 4.0, 4.1, 4.1, 4.3, 3.8

Lower Quartile:

Upper Quartile:

Interquartile Range:

4

The following are the scores Elizabeth earned on her last 7 math assessments

82, 97, 100, 94, 89, 89, 92

What is the IQR of her scores?

5

Michaela completed 4 timed 50- meter freestyle laps. Her times are recorded below:

48.5, 43.8, 42.5, 47.6

What is the IQR of her times?

6

A police officer used his radar gun to record the speeds of passing cars. The speeds he recorded are listed below:

57, 65, 70, 65, 72, 62

What is the IQR of the car speeds?

7

8 students in Mrs. Morris's class completed a timed math facts drill in 1 minute. The number of problems each student completed is recorded below:

19, 25, 32, 48, 35, 30, 35, 40

What is the IQR of the students' results?

8

Mrs. Colton measured the feet of 11 students in her class. Their shoe sizes are listed below:

6.5, 7, 8, 6.5, 5, 6, 6, 8, 6.5, 9, 7.5

What is the IQR of the shoe sizes?

9

The table below shows the average monthly precipitation (in) for the U.S.

Month	Jan	Feb	Mar	Apr	May
Amount (in inches)	2.7	2.5	2.9	3.8	3.2

What is the IQR of the U.S. precipitation data?

10

The table below shows the average monthly temperature (°F) for the North Carolina

Month	Jan	Feb	Mar	Apr	May
Temp (°F)	40	42.2	49.7	58.1	66.1

What is the IQR of the North Carolina temperature data?

11

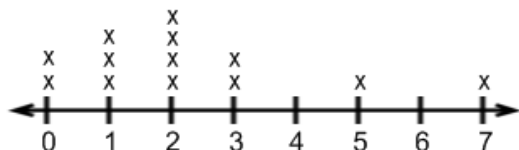
The table below shows how many points the Eagles basketball team scored each game

What is the IQR of the Eagles' scores?

Game	Points Scored
1	33
2	24
3	48
4	37
5	45
6	34
7	65
8	32

12

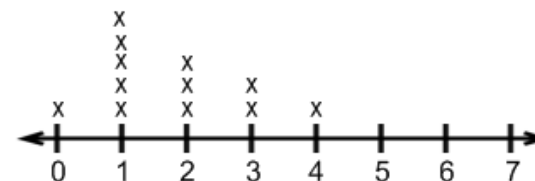
The line plot shows the number of pets students own in Ms. Flowers' class.



What is the IQR of the # of pets students own in Ms. Flowers' class?

13

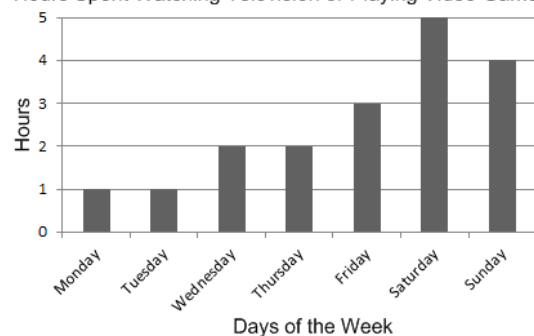
The line plot shows the number of siblings students in Mrs. Colton's class have.



What is the IQR of the # of siblings students in Mrs. Colton's class have?

14

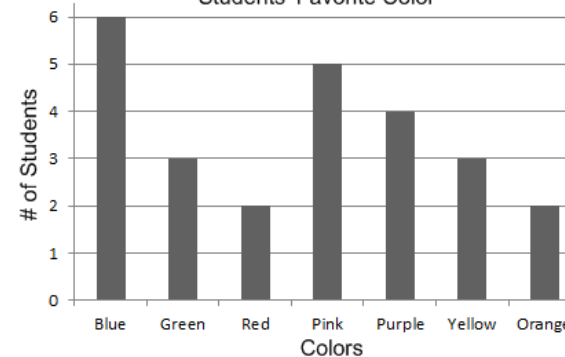
Hours Spent Watching Television or Playing Video Games



What is the IQR of the amount of time students spend using electronics?

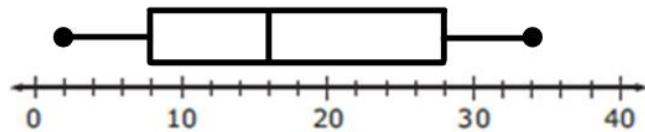
15

Students' Favorite Color



What is the IQR of the students' favorite color?

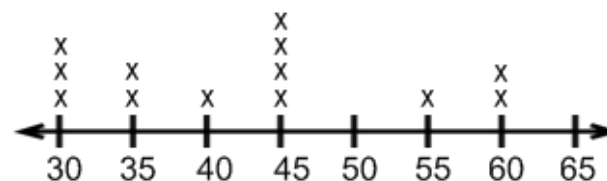
16



According to the box and whisker plot, what is the interquartile range (IQR)?

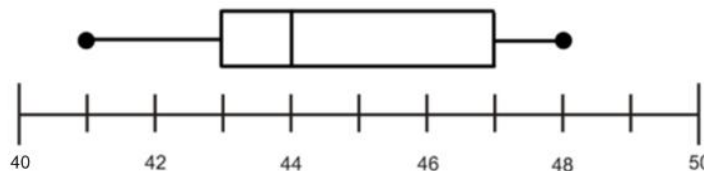
17

The line plot shows the number of minutes patrons at a local gym spend exercising every workout.



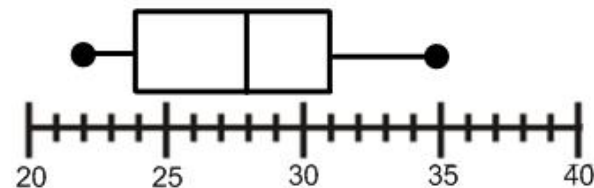
What is the IQR of the amount of time patrons spend exercising?

18



According to the box and whisker plot, what is the interquartile range (IQR)?

19



According to the box and whisker plot, what is the interquartile range (IQR)?

20

1. Lower Quartile: Upper Quartile: IQR:	2. Lower Quartile: Upper Quartile: IQR:
3. Lower Quartile: Upper Quartile: IQR:	4. Lower Quartile: Upper Quartile: IQR:
5.	6.
7.	8.
9.	10.
11.	12.
13.	14.
15.	16.
17.	18.
19.	20.

ANSWER KEY

When you find the Interquartile Range (IQR) of a set of data, you need 3 data points.

1. The median is the middle of the data. Make sure you put your data in order from least to greatest first! The median helps you cut your data into the lower and upper quartiles.
2. The lower quartile (or quartile 1) is found by finding the median of the lower half of the data.
3. The upper quartile (or quartile 3) is found by finding the median of the upper half of the data.

Finding the IQR with an Odd # of Data Points

1. Put the numbers in order from least to greatest and find the median of the data. Circle it.
2. Find the median of the lower quartile (the numbers in front of your median).
3. Find the median of the upper quartile (the numbers behind your median).

(Hint: If a quartile contains an even # of data points, add the two middle numbers and divide by 2 to get your median.)

4. Subtract the lower quartile from the upper quartile ($Q3 - Q1$).

Finding the IQR with an Even # of Data Points

1. Put the numbers in order from least to greatest.
2. Break your data into two equal groups. Draw a line between the two groups.
3. Find the median of the front group. (lower quartile)
4. Find the median of the back group. (upper quartile)
5. Subtract the lower quartile from the upper quartile ($Q3 - Q1$).

ANSWER KEY

Let's Try it! – Remember: Median, Median, Median, Subtract!

Finding the IQR with an Odd # of Data Points

A rescue agency recorded the number of pets adopted each month:

12, 18, 23, 15, 29, 12, 8, 17, 16

What is the IQR of the # of pets adopted at the rescue agency?

1. Put the numbers in order.

8, 12, 12, 15, 16, 17, 18, 22, 29

2. What is your median? Circle it.
3. Find the median of the numbers in front of your median (Q1).
There are 4 numbers, but the two in the middle are the same, so that is your median, or Q1. $Q1 = 12$
4. Find the median of the numbers behind your median (Q3). There are four numbers. The two in the middle are not the same, so add them up and divide by 2. $Q3 = \frac{18 + 22}{2} = 20$
5. Now $Q3 - Q1 = 20 - 12 = 8$

Let's Try Another!

Find the IQR of the data below:

23, 18, 25, 21, 21, 17, 12
12, 17, 18, 21, 21, 23, 25

Lower Quartile: **17.5**

Upper Quartile: **22**

Interquartile Range: $22 - 17.5 = 4.5$

Finding the IQR with an Even # of Data Points

A car rental agency rented a certain number of cars each day::

25, 12, 15, 11, 18, 20

What is the IQR of the # of cars rented at the agency?

1. Put the numbers in order.

11, 12, 15, 18, 20, 25

2. Cut the #s into two equal groups. Draw a line to separate them.
3. Find the median of the numbers in the front half – this is Q1. $Q1 = 12$
4. Find the median of the numbers in the back half – this is $Q3 = 20$
5. Now $Q3 - Q1 = 20 - 12 = 8$

Let's Try Another!

Find the IQR of the data below:

21, 17, 15, 23, 21, 17, 12, 9

9, 12, 15, 17, 17, 21, 21, 23

Lower Quartile: **13.5**

Upper Quartile: **21**

Interquartile Range: **7.5**

1. Lower Quartile: 19.5 Upper Quartile: 24 IQR: 4.5	2. Lower Quartile: 15 Upper Quartile: 14 IQR: 9
3. Lower Quartile: 18 Upper Quartile: 23 IQR: 8	4. Lower Quartile: 3.9 Upper Quartile: 4.3 IQR: 0.4
5. $97 \div 89 = 8$	6. $48.05 - 43.15 = 4.9$
7. $70 \div 62 = 8$	8. $37.5 - 27.5 = 10$
9. $8 \div 6 = 2$	10. $3.5 - 2.6 = 0.9$
11. $62.1 - 41.1 = 21$	12. $48 \div 32 = 16$
13. $3 \div 1 = 2$	14. $2.5 - 1 = 1.5$
15. $4 \div 1 = 3$	16. $5 \div 2 = 3$
17. $28 \div 8 = 10$	18. $50 - 32.5 = 17.5$
19. $47 \div 43 = 4$	20. $31 \div 24 = 7$